

Ayush Yadav

☎ +91 9236593446 ✉ ayushthe8055@gmail.com [🌐 LinkedIn](#) [🐙 GitHub](#)

EDUCATION

Indian Institute of Technology, Patna

Bachelor of Technology in Computer Science and Engineering

August 2024 – May 2028

CPI: 8.71

Lucknow Public School, Lucknow, UP

Central Board of Secondary Education

April 2012 – May 2023

Percentage: 96.6%

COMPETITIVE PROGRAMMING

- **Specialist** on **Codeforces** with peak rating **1470** [Profile Link](#)
- **Knight** on **LeetCode** with peak rating **1886** [Profile Link](#)
- **3 Star Coder** on **Codechef** with peak rating **1624** [Profile Link](#)
- Solved over **1400+** algorithmic problems across **Codeforces**, **LeetCode**, **CodeChef**, and **AtCoder**.

ACHIEVEMENTS

- **ICPC Prelims 2025:** Secured **Rank 384**, solved **4 problems**, Team: **XNOR** (Team ID: **icpcpr251399**)
- Secured an **All India Rank 3941** out of **1.4 million+** candidates in **JEE Main 2024**
- Secured an **All India Rank 5581** out of **180,000** candidates in **JEE Advanced 2024**

PROJECTS

Two-Pass Assembler and Emulator | *C, GCC* | [Source Code](#)

- Developed a two-pass assembler that translates a custom assembly language into 32-bit machine code while generating object, listing, and log files.
- Implemented symbol table management, label resolution, assembler directives (**data**, **SET**), and comprehensive syntax and semantic error detection.
- Built a machine emulator supporting instruction execution, trace generation, memory dumps, and runtime error handling including illegal opcodes and infinite loop detection.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL

Frameworks/Technologies: React.js, Node.js, Express.js, MongoDB

Tools/Platforms: Git, GitHub, MongoDB Atlas, $\text{\LaTeX}$

RELEVANT COURSEWORK

Data Structures and Algorithms, Computer Architecture, Database Management Systems, Discrete Mathematics, Linear Algebra, Statistics and Probability